

Quick overview

Version: 3.8 (stable release)

Licence: GNU general-public licence version 2.0 (GPLv2)

Supported OS: Mac OS X, RedHat Linux, OpenBSD, Slackware 8 Linux, Ubuntu Linux, Pardus Linux, Windows 7 32-bit and 64-bit

Top three features:

1. Easy access to editing tools and library with a toolbar on the right
2. Message window to indicate the cursor position along with the version and revision number of the tool
3. Schematic-to-symbol representation switch with the press of buttons at the bottom

Note: The latest version of XCircuit is 3.9, which is a development release.

Benefits of hierarchical design

A hierarchical design has a number of benefits associated with it.

1. Replicating a block and making changes at block-level becomes easy.
2. Connectivity derived from blocks and ports enables name-conflict issues.
3. Captures constraints even when logic is being defined.
4. Reuse of functional blocks across multiple designs becomes easy as compared to the ones from flat schematics.

edit the output and view its source in detail. Being an interpreted language, PostScript helps XCircuit to create a set of definitions that can help it overcome a set of issues related to circuit drawing. The expansion of the set makes these compatible with earlier versions.

Although PostScript is a printer language, it saves files in the same format that is used to print these. Thus, printing output designs with XCircuit is not that tough.

From 2D architectural designs to PCBs

As discussed earlier, XCircuit can be used as a generic program to draw 2D designs and circuit architecture. You can design the architectural drawing, music typography, printed circuit boards (PCBs) and almost anything that requires the use of repeated standard set of graphical objects.

Create your own library. Although the main aim of XCircuit is to create circuit diagrams, it can also be used to create symbols that can be saved in a library and used later during circuit design. Tim Edwards, who wrote XCircuit, says, "I created an example symbol in a library in XCircuit. I can now use that symbol as part of any drawing in XCircuit if I wanted to. I think XCircuit would be

a good choice for arbitrary diagram drawing needs."

Easy LaTeX embedding. PostScript (.ps) format of files saved in XCircuit allows you to embed the circuit in LaTeX document easily. As LaTeX works on TeXnicCenter, you only need to open the document and select Insert Picture to use the saved file.

Recover files on crash. XCircuit has been created in a way that a copy of its current work is saved in the temporary (/tmp) directory, which is removed when you exit the program.

However, if the program shuts down unexpectedly, or if the system crashes, the current work remains in the /tmp directory.

Now, you open XCircuit again. It will prompt you to recover the file in the same way as Microsoft Office or Open Office. However, this feature has a small shortcoming. XCircuit creates a backup of files periodically, once in every ten minutes. Thus, changes made in the last few minutes of the crash do not get updated.

Default single-click. Key and button bindings in XCircuit use the name Hold to indicate that these can be operated by pressing a key or button. It does not use the double-clicking option by default. However, the tool command language tcl/tk version of XCircuit makes use of tk Bind command to allow multiple clicks (double, triple and quadruple). These options are not available with default button bindings.

Quick library messages. XCircuit has another message window at the bottom of its main screen to provide quick information regarding the library that has just loaded (for example, /usr/local/lib/XCircuit-3.2/analoglib2.lps). In the non-tcl version of XCircuit, this message disappears after ten seconds. The message win-

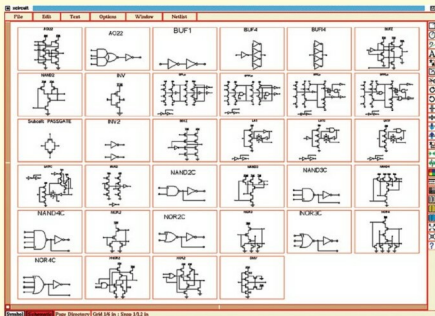


Fig. 3: Front-end schematic